Fluid Mechanics
Project based Learning

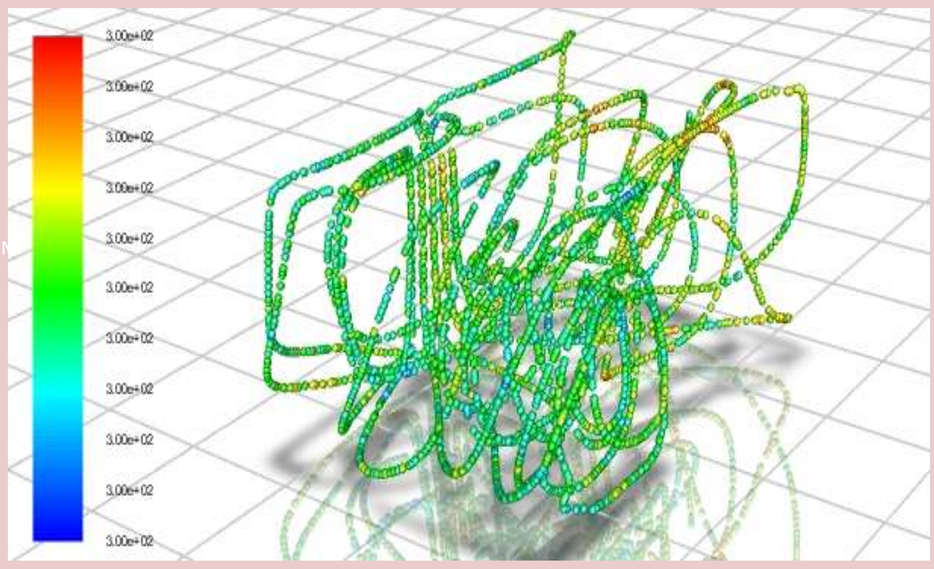
Topic: AIR CONDITIONER IN A ROOM

Introduction

This m flow in a room.
Using ANSYS FLUENT, analyze the flow in a room with air conditioner
The science which manages making a controlled in indoor space is alluded to as cooling. Prior days the cooling wind up regarded
as a luxurious, anyway in gift situation because of substitute in natural conditions the cooling framework has risen as an integral
part of human ways of life. In tropical and subtropical nations cooling by utilizing cooling is fundamental capacities of front-line
advancement. In this manner, it is important to create present day cooling framework to meet the necessities comfort conditions

Results

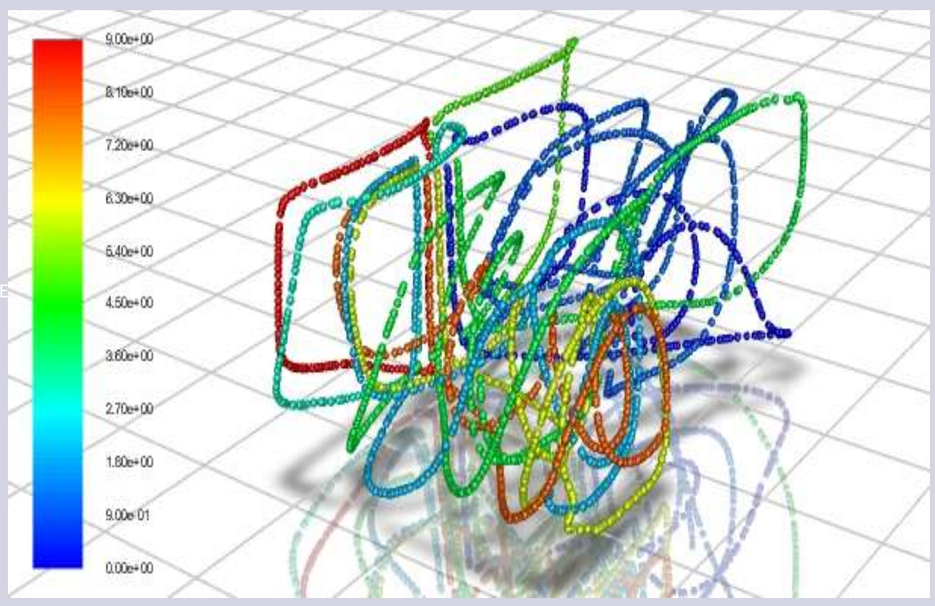
PATHLINES



Objective

Increased Fuel consumption now-a-days is a great concern for customers as well as for carmakers. To overcome this problem carmakers often try to improve the aerodynamics of the car so that fuel consumption can be minimized. Aerodynamics can be improved by making design changes to the car or adding some aerodynamics components like spoilers, roof rails or vortex generator to the car. Components like vortex generators and spoilers offers great drag resistance and reduce the lift of the car. They also increases the mileage of a car.
To increase fuel efficiency and performance to greater extent some design changes can be made to the existing designs of Spoilers and vortex generator.
Shapes can be modified to reduce the mass and to increase aerodynamics more effectively so that fuel consumption can decrease to a good extent even at higher speeds.

Results

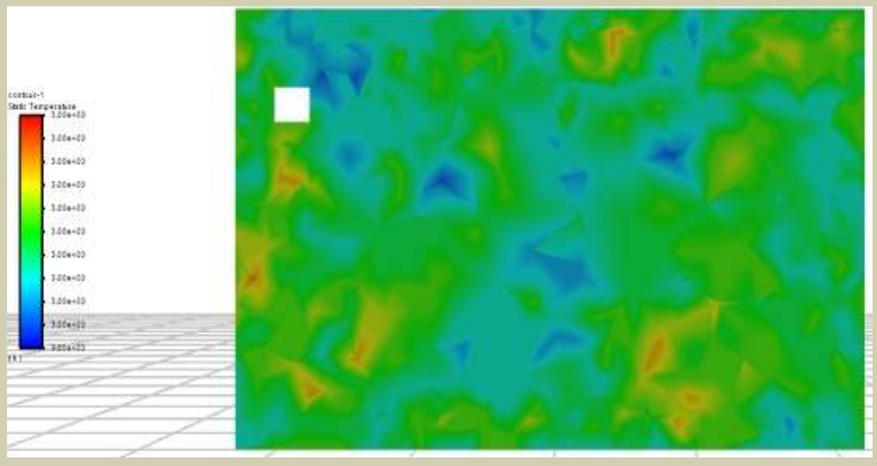


Physics set up

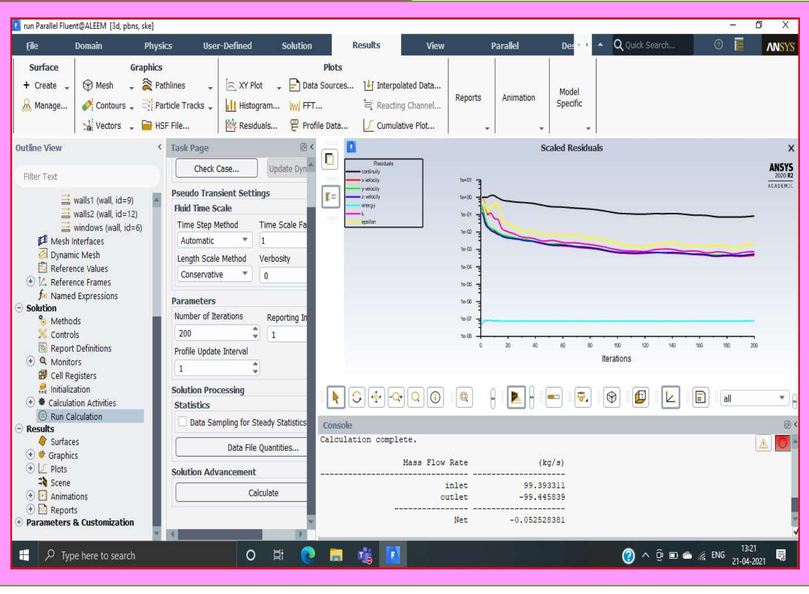
*Check mesh.
*Display mesh.
*Use steady state pressure based solver.
*Make energy ON. Set viscous model. Use k-epsilon model.
*Set material properties. Use defaults.
*Define cell zone conditions.
*Define boundary conditions. Set inlet as Velocity-Inlet with Velocity Magnitude of 3 m/s.Set Turbulence Specification Method to Intensity and Viscosity Ratio with values of 1 and 5 resp. Set Temperature to 283K. Set outlet zone type to Pressure-Outlet. Set Turbulence Specification Method to Intensity and Viscosity Ratio with values of 1 and 5 resp. Set Temperature to 300K. For zone walls1, set Convection Thermal Condition with HTC as 5 w/m2-k.
*Define operating conditions. Set Gravity as -9.81 m/s2 in Z direction.

Results

TEMPERATURE CONTOUR

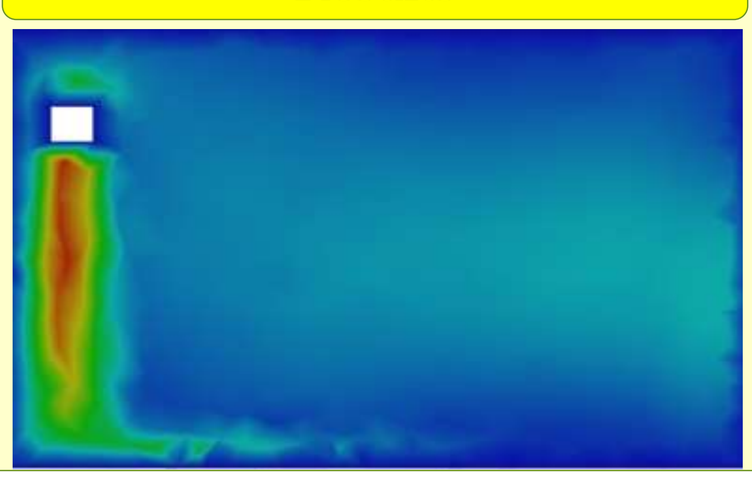


Numerical solution



Results

VELOCITY CONTOUR



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